

Abstract

Developing a Culturally Adapted Adolescent Life-Skills Curriculum (THRIVE-Belize) Using Community-Based Participatory Research and AI-Assisted Synthesis

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Background

A school-based survey (N=509) at Toledo Community College (TCC) in southern Belize found that 32.1% of students aged 12-17 reported self-harm, 26.2% reported a suicide attempt, 54.7% had witnessed neighborhood violence, and 38.3% had experienced interparental violence. National data identified Toledo District adolescents as the subgroup with the highest violence-endorsing attitudes in Belize. No culturally adapted life-skills curriculum existed to address these intersecting risks.

Methods

Using community-based participatory research (CBPR), we developed the THRIVE-Belize facilitator manual with TCC stakeholders and Hillside Health Care (local NGO). A multi-agent artificial intelligence (AI) system with human-in-the-loop oversight synthesized 18 evidence-based source curricula, selected activities by compliance, feasibility, and skill-teaching criteria, and adapted content for Belize's multi-ethnic adolescent population (Kriol, Maya, Mestizo, Garifuna). An explicit AI governance framework enforced ethical constraints, quality gates, and human review at every decision point.

Results

The process produced a 31-session, 7-module life-skills curriculum addressing communication, gender norms, sexual and reproductive health, healthy relationships, mental health, substance use, and environmental health. The manual contains 95 activities—80 adapted from source curricula and 15 generated to fill identified gaps—with 78 documented adaptations (22 cultural localizations), a structured audit trail, and a complete evidence map tracing each activity to its source. All sessions use 40-minute, paper-based, low-cost formats suitable for low-resource school settings.

Conclusions

Integrating CBPR with AI-assisted curriculum development offers a replicable, transparent, and culturally responsive approach to creating adolescent health interventions for underserved populations. Phase 1 feasibility and acceptability testing at TCC is planned.

Assessment of individual and community needs for health education Communication and informatics Diversity and culture Planning of health education strategies, interventions, and programs Public health or related research Social and behavioral sciences

